Introduction to Database Management Systems

Overview of Database Systems

Topic Objectives

- History of Database Systems
- Know the potential problems with lists
- Understanding the reasons for using a database
- Understanding how related tables avoid the problems associated with lists
- Learn the components of a database system
- Learn the elements of a database
- Learn the purpose of a database management system (DBMS)
- Understanding the functions of a database application

History of Database Systems

- First-generation
  - Hierarchical and Network
- Second generation
  - Relational
- Third generation
  - Object-Relational
  - Object-Oriented

Purpose of a Database

- The purpose of a database is:
  - To store data
  - To provide an organizational structure for data
  - To provide a mechanism for querying, creating, modifying, and deleting data
- A database can store information and relationships that are more complicated than a simple list

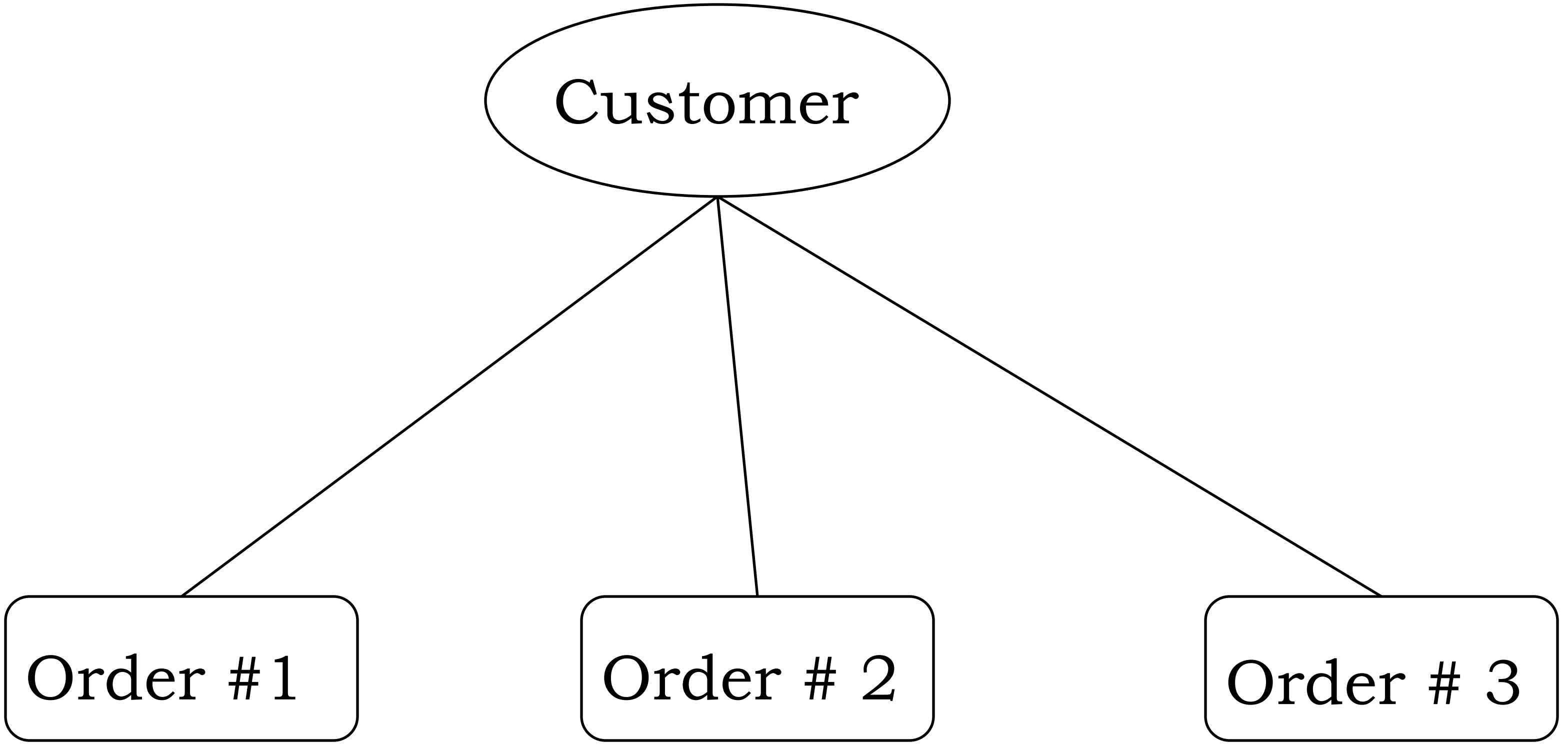
Create =	C
Read =	R
Update =	U
Delete =	D

Create , Read, Update , Delete = (CRUD)
These are the four basic operation when interacting with the database

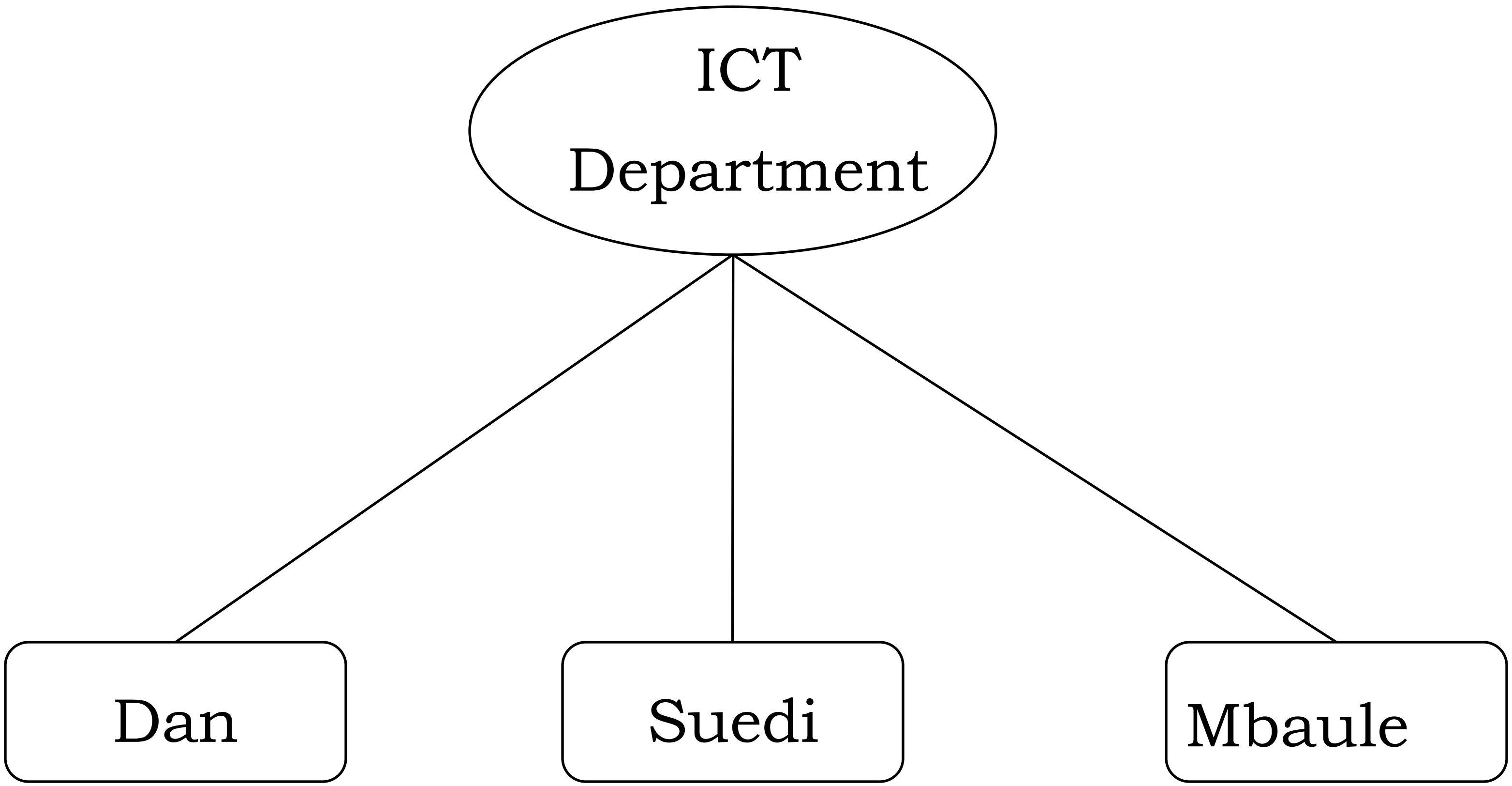
What is a Database?

- A database:
  - is a organized **collection** of **related data** and **Data** is a collection of recordable **facts** and **figures** that can be digitally processed based on some **business logic.** For example, a university database might contain information about the following:
    - ***Entities*** *such as students, faculty, courses, and classrooms.*
    - ***Relationships between entities***, *such as students' enrollment in courses, faculty* teaching courses, and the use of rooms for courses.
- A database captures an abstract representation of the domain of an application.
  - Typically organized as “records” (traditionally, large numbers, on disk) and
  - relationships between records

Relationship in Database



Relationship in Database



Problem with Lists: Redundancy

- In a list, each row is intended to stand on its own. As a result, the same information may be entered several times
  - For example, a list of projects may include the project manager's name, ID, and phone extension.
    - If a particular person is currently managing 10 projects, his/her information would appear in the list 10 times

Problem with Lists: Redundancy

Project Manager	Manager ID	Phone
Dan	3	0754xxx
Dan	3	0754xxx
Dan	3	0754xxx
Eliza	2	0629xxx
Dan	3	0754xxx
Dan	3	0754xxx
Kikali	1	0713xxx
Dan	3	0754xxx
Dan	3	0754xxx
Kikali	1	0713xxx
Dan	3	0754xxx

Problem with Lists: Redundancy

Project Manager	Manager ID	Phone	Project Name	Project ID	Start Date	Budget
Dan	3	0754xx	OS Upgrade	101	01 Mar 2021	22,000k
Dan	3	0754xx	SQL training	102	03 Apr 2021	12,000k
Dan	3	0754xx	Signal Boost	103	20 Aug 2021	10,000k
Eliza	2	0629xx	Website design	104	30 Oct 2021	22,000k
Dan	3	0754xx	Cloud hosting	105	11Nov 2021	32,000k
Dan	3	0754xx	A.I. Research	106	20Jan 2022	63,000k
Kikali	1	0713xx	Web Server	107	20 Feb 2022	13,000k
Dan	3	0754xx	Client training	108	20 Mar 2022	33,000k
Dan	3	0754xx	Python training	109	20 Apr 2022	43,000k
Kikali	1	0713xx	3D Printers	110	24 May 2022	23,000k
Dan	3	0754xx	DB upgrade	112	14 Jun 2022	12,000k

List Modification Issues

- Redundancy and multiple themes in lists create modification problems
  - Deletion problems
  - Update problems
  - Insertion problems

List Modification Issues

If Adviser **Baker** is changed to **Taing**, we need to change *AdviserEmail* as well. If changed to **Valdez**, we need to change *AdviserEmail*, *Department*, and *AdminLastName*.

	A	B	C	D	E	F	G
1	LastName	FirstName	Email	AdviserLastName	AdviserEmail	Department	AdminLastName
2	Andrews	Matthew	Matthew.Andrews@ourcampus.edu	Baker	Linda.Baker@ourcampus.edu	Accounting	Smith
3	Brisbon	Lisa	Lisa.Brisbon@ourcampus.edu	Valdez	Richard.Valdez@ourcampus.edu	Chemistry	Chaplin
4	Fischer	Douglas	Douglas.Fischer@ourcampus.edu	Baker	Linda.Baker@ourcampus.edu	Accounting	Smith
5	Hwang	Terry	Terry.Hwang@ourcampus.edu	Taing	Susan.Taing@ourcampus.edu	Accounting	Smith
6	Lai	Tzu	Tzu.Lai@ourcampus.edu	Valdez	Richard.Valdez@ourcampus.edu	Chemistry	Chaplin
7	~~Marino~~	~~Chip~~	~~Chip.Marino@ourcampus.edu~~	~~Tran~~	~~Ken.Tran@ourcampus.edu~~	~~InfoSystems~~	~~Rogers~~
8	Thompson	James	James.Thompson@ourcampus.edu	Taing	Susan.Taing@ourcampus.edu	Accounting	Smith
9	???	???	???	???	???	Biology	Kelly

Deleted row—Student, Adviser, and Department data lost

Inserted row—both Student and Adviser data missing

Addressing Information Complexities

Relational databases are designed to address many of the information complexity issues that arise in business

Relational databases

- A relational database stores information in tables. Each information them (business concept) is stored in its own table.

- In essence, a relational database will break- up a list into a several parts.

- One part for each theme in the list
- For example, a Project List might be divided into a CUSTOMER Table, a PROJECT Table, and a PROJECT_MANAGER Table

A Table

A table has two dimensional

i. Column- represents a different attribute of an entities

ii. Rows – Each row of table represents instance of the entities

Relational databases

Employee ID	**Name**	**Phone number**	**Department ID**
101	Dan	714-555-7270	1
102	Br. James	657-555-6561	2
103	Br. Pius	303-555-3247	2
104	Boniface	480-555-4439	2

Department ID	Department Name
1	Information Systems
2	Package delivery

Putting the Pieces Back Together

- In our relational database example, we broke apart our list into several tables. Somehow the tables must be joined back together
- In a relational database, tables are ***joined*** together using matched pairs of data values

- Fore example, if a PROJECT has a CUSTOEMR, the customer_ID can be stored as a column in the Project Table. Whenever we need information about a customer, we can use the customer_ID to look up the customer information in the customer table

Project Name	Project ID	Start Date	Budget	Customer ID
Web Server	107	20 Feb 2022	13,000k	2
Client training	108	20 Mar 2022	33,000k	2
Python training	109	20 Apr 2022	43,000k	1
3D Printers	110	24 May 2022	23,000k	2
DB upgrade	112	14 Jun 2022	12,000k	1

Department ID	Department Name
1	Dan
2	Swedi

Sounds like More Work, Not Less

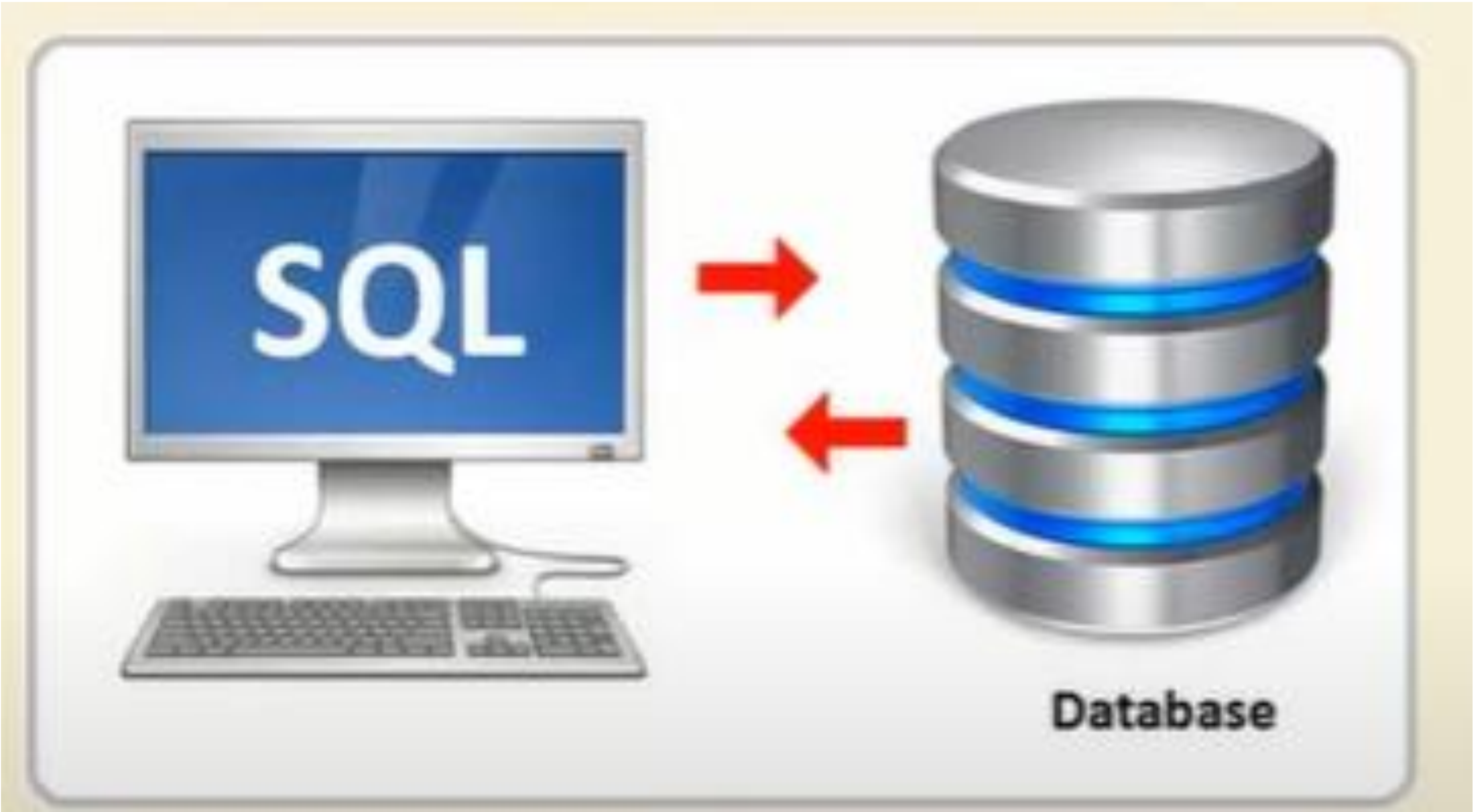
- A relational database is more complicated than a list
- However, a relational database minimizes data redundancy, preserves complex relationships among topics, and allows for partial data (*null* values)
- Furthermore, a relational database provides a solid foundation for creating user interface forms and reports

The structured Query Language (SQL)

The structured Query Language (SQL) is an international standard language for creating, processing, and querying databases and their tables.

The vast majority of data-driven applications and websites use SQL to retrieve, format, report, insert, delete and/or modify data for users.

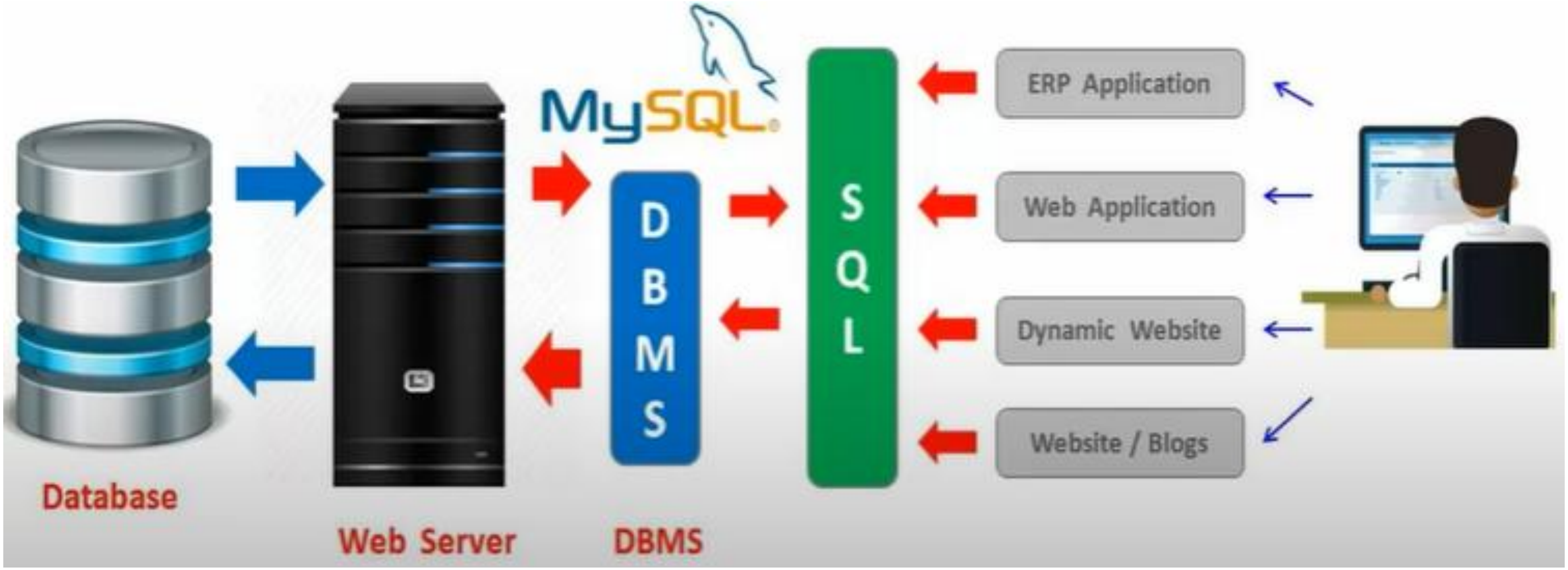
SQL - Structured Query Language



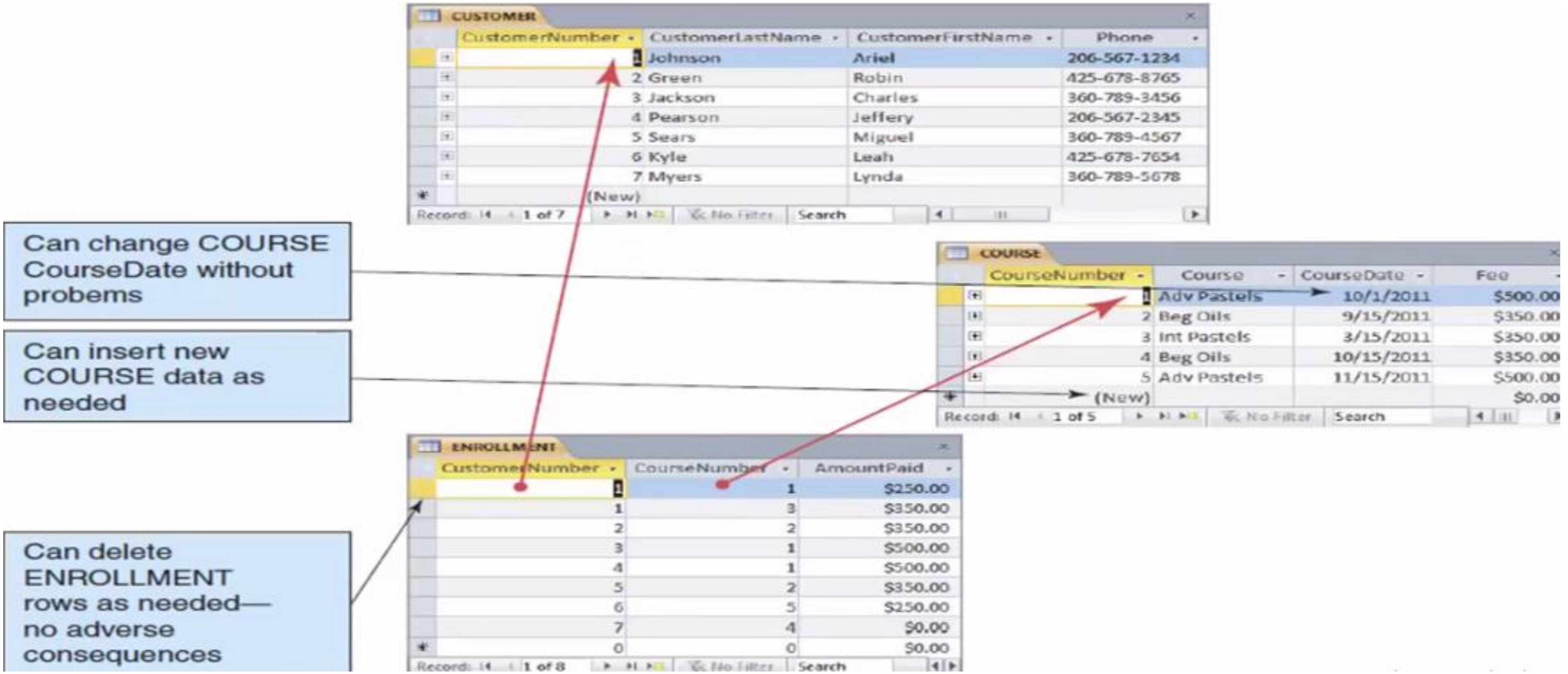
SQL is used to performs database operation such as Create, Update, Delete (CRUD) object such as database, table records, field.

Common used in SQL

SQL - Structured Query Language



SQL Example: The Art Course Database



CUSTOMER

CustomerNumber	CustomerLastName	CustomerFirstName	Phone
1	Johnson	Ariel	206-567-1234
2	Green	Robin	425-678-8765
3	Jackson	Charles	360-789-3456
4	Pearson	Jeffery	206-567-2345
5	Sears	Miguel	360-789-4567
6	Kyle	Leah	425-678-7654
7	Myers	Lynda	360-789-5678
(New)			

Record: 1 of 7 No Filter Search

COURSE

CourseNumber	Course	CourseDate	Fee
1	Adv Pastels	10/1/2011	$500.00
2	Beg Oils	9/15/2011	$350.00
3	Int Pastels	3/15/2011	$350.00
4	Beg Oils	10/15/2011	$350.00
5	Adv Pastels	11/15/2011	$500.00
(New)			$0.00

Record: 1 of 5 No Filter Search

ENROLLMENT

CustomerNumber	CourseNumber	AmountPaid
1	1	$250.00
1	3	$350.00
2	2	$350.00
3	1	$500.00
4	1	$500.00
5	2	$350.00
6	5	$250.00
7	4	$0.00
0	0	$0.00

Record: 1 of 8 No Filter Search

SQL Example

- We can use SQL to combine the data in the three tables in the Art Course Database to recreate the original list structure of the data.

- We do this by using a SQL ***SELECT*** statement

SQL Example (Continued)

- SELECT Customer.CustomerLastName,
  Customer.CustomerFirstName,
  Customer.phone,
  Customer.Coursedate, Enrollment.amountPaid,
  Customer.Course, Course.fee

- FROM Customer, Enrollment, Course

- WHERE Customer.customerNumber = Enrollment. customerNumber

- AND Course.courseNumber = Enrollment.courseNumber

SQL Example Results

Art Course List

CustomerLastName	CustomerFirstName	Phone	CourseDate	AmountPaid	Course	Fee
Johnson	Ariel	206-567-1234	10/1/2011	$250.00	Adv Pastels	$500.00
Johnson	Ariel	206-567-1234	3/15/2011	$350.00	Int Pastels	$350.00
Green	Robin	425-678-8765	9/15/2011	$350.00	Beg Oils	$350.00
Jackson	Charles	360-789-3456	10/1/2011	$500.00	Adv Pastels	$500.00
Pearson	Jeffery	206-567-2345	10/1/2011	$500.00	Adv Pastels	$500.00
Sears	Miguel	360-789-4567	9/15/2011	$350.00	Beg Oils	$350.00
Kyle	Leah	425-678-7654	11/15/2011	$250.00	Adv Pastels	$500.00
Myers	Lynda	360-789-5678	10/15/2011	$0.00	Beg Oils	$350.00

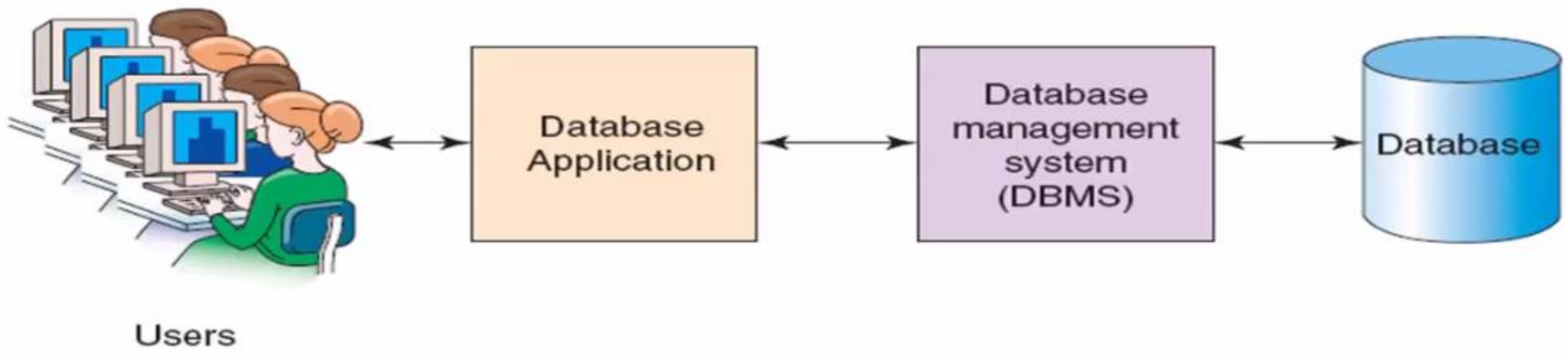
Record: 1 of 8 No Filter Search

Database Systems

- The four components of a database system are:

    1. Users
    2. Database Application(s)
    3. Database Management System (DBMS)
    4. Database

Components of a Database Systems



Users

- A user of a database system will:
  - Use a database application to keep track of information
  - Use different user interface forms to enter, read, delete, and query data
  - Produce reports

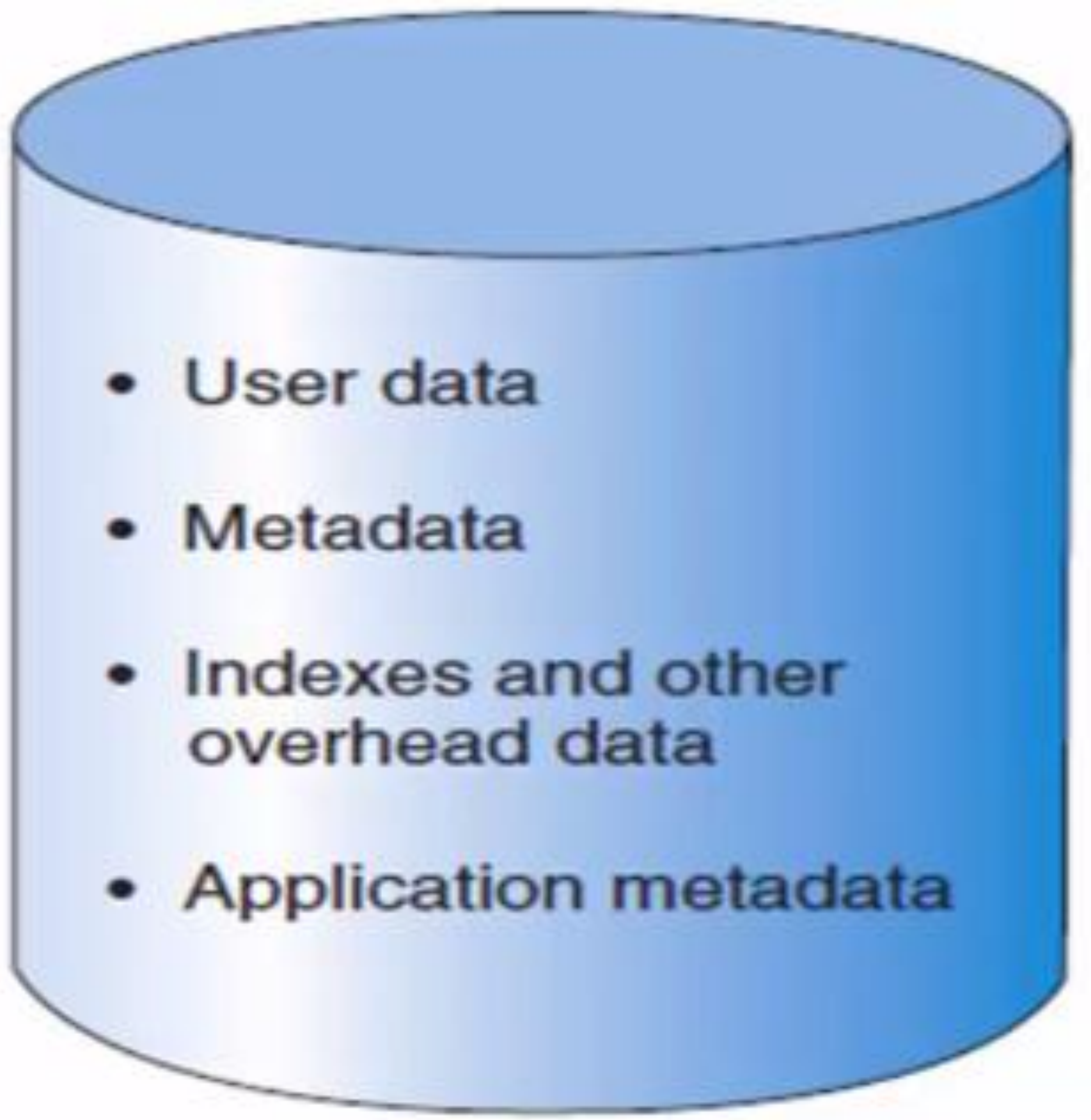
The Database

- A database is a *self-describing* collection of *related* records
- Self-describing:
    - The database itself contains the definition of its structure
    - Metadata are data describing the structure of the data in the database
- Tables within a relational database are related to each other in some way.

Example of Metadata

Comment ID	Name	Elements
1	Tree	Very low
2	Pothole	Low
3	Gravel	Medium
4	Cable stone	High

Database contents



Database Management System (DBMS)

- A database management system (DBMS) serves as an *intermediary* between database applications and the database
- The DBMS manages and controls database activities
- The DBMS creates, processes, and administers the databases it controls

Functions of a DBMS

- Create databases
- Create tables
- Create supporting structures
- Read database data
- Modify database data (Insert, update, delete)
- Maintain database structures
- Enforce rules
- Control concurrency
- Provide security
- Perform data backup and recovery

Referential Integrity Constraints

- A DBMS can enforce many constraints…
- Referential integrity constraints ensure that the values of a column in one table are valid based on the values in another table
  - For example, if a 5 was entered as a CustomerID in the PROJECT table, a Customer having a CustomerID value of 5 <u>must</u> exist in the CUSTOMER table

An Example

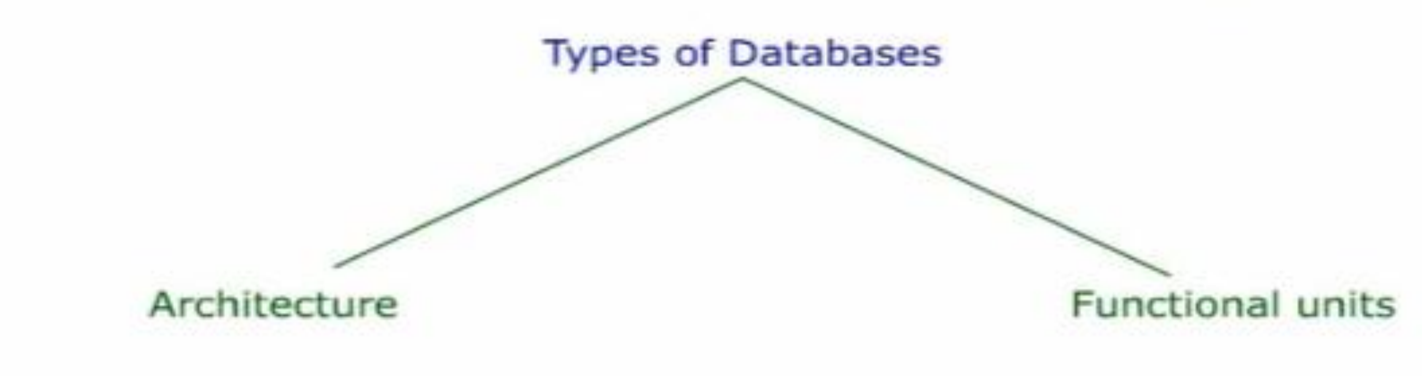
Project Name	Project ID	Start Date	Budget	Customer ID
DB upgrade	110	19 Nov 2015	$8700	2
New email server	111	09 Jan 2016	$11900	2
3D printers	112	06 Jul 2016	$25000	1
Network upgrade	113	24 Aug 2016	$33200	5

Customer ID	Customer Name
1	Dan
2	Priya

Database Application(s)

- A database application is a set of one or more computer programs or websites that serve as intermediary between the user and the DBMS

Types of Database



Architecture

i. Personal database systems
ii. Enterprise-level Database System
iii. Client Server database
iv. Distributed database

Functional Units

i. Personal Computer database
ii. Workgroup database

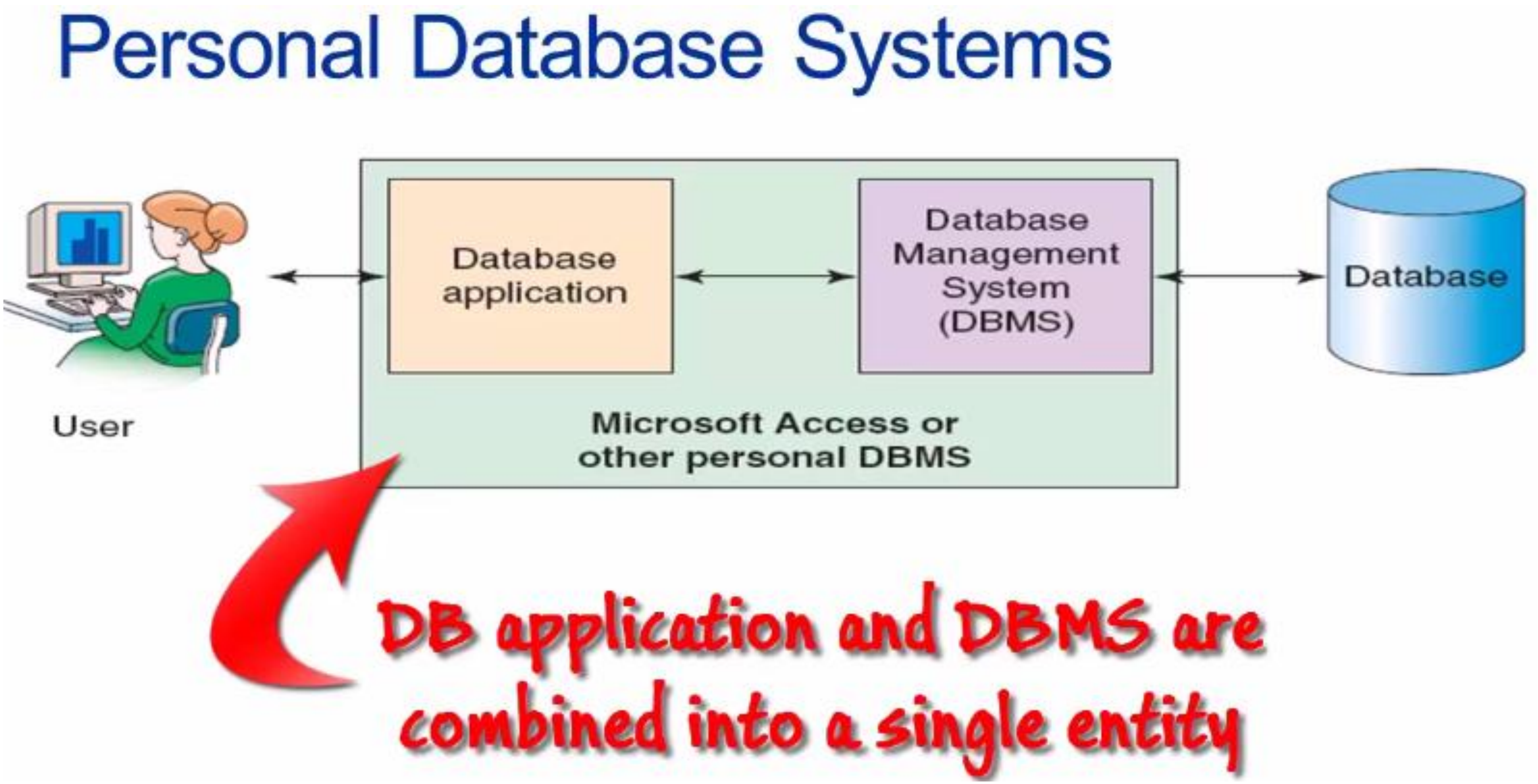
Types of Database

Personal Database Systems

Characteristics

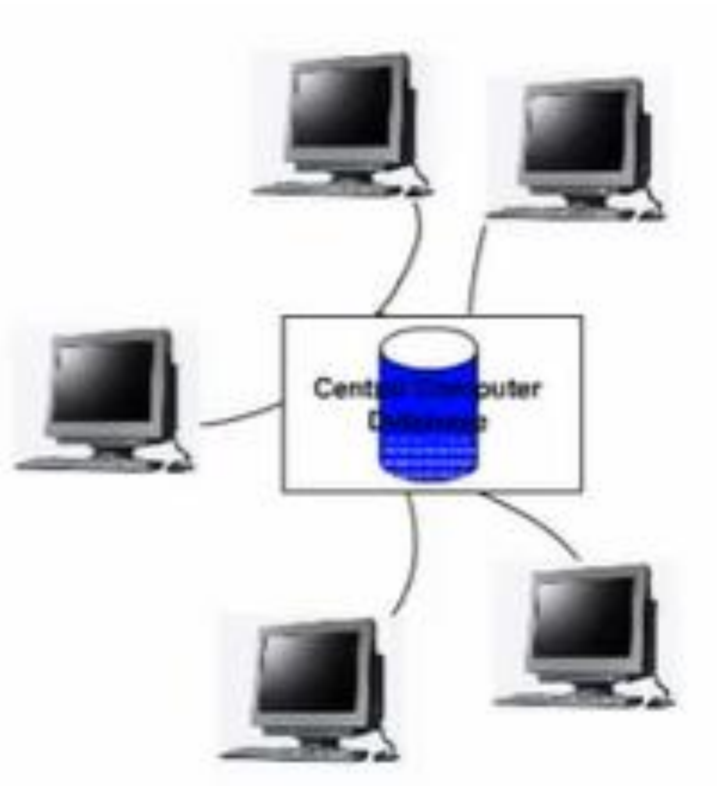
- Support one application
- Have only a few tables
- Are simple in design
- Involve only one computer
- Support one user at a time

An Example Personal Database Systems



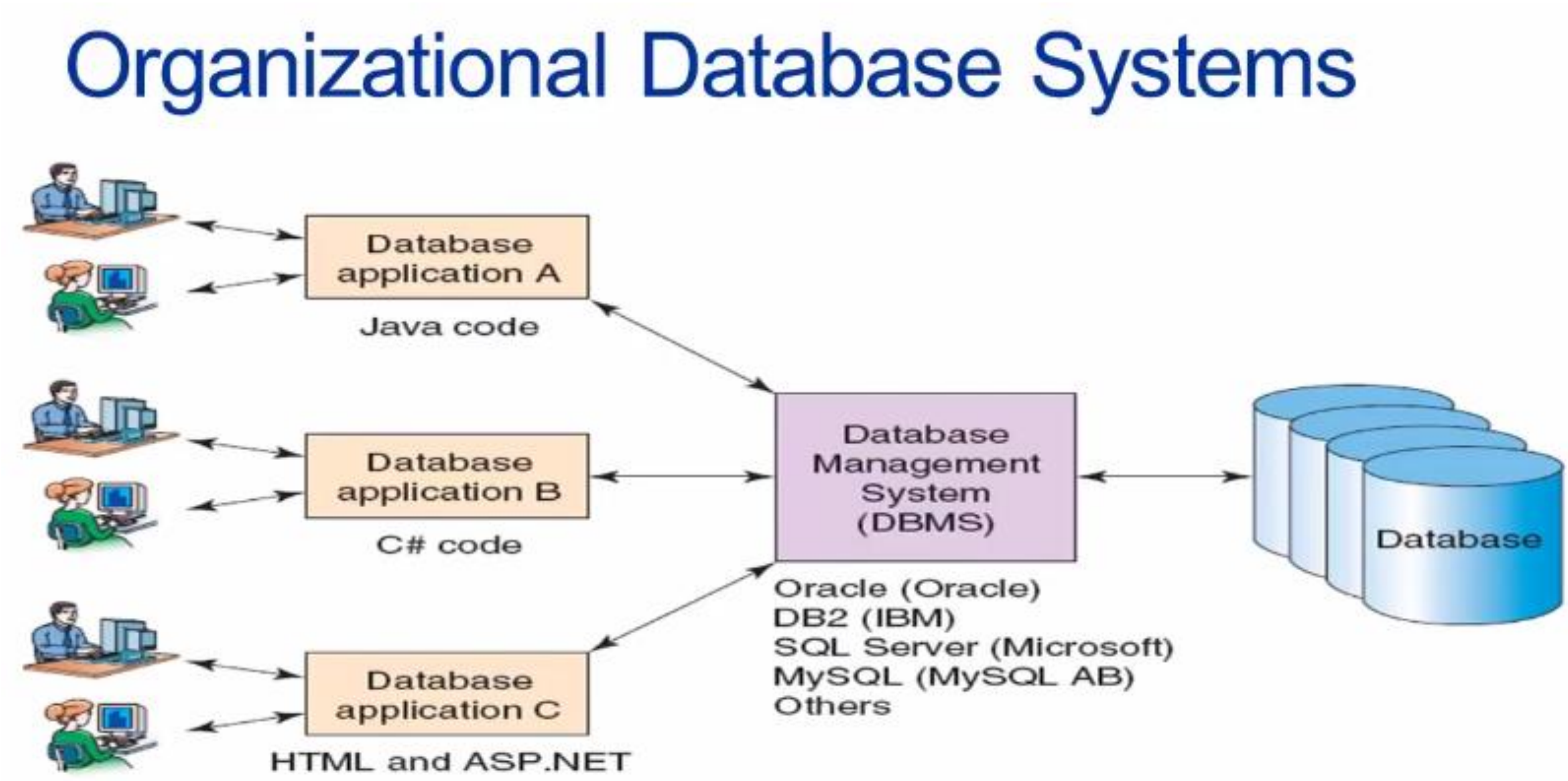
Types of Database contd….

Enterprise-level Database System



- Support several users simultaneously
- Support more than one application
- Involve multiple computers
- Are complex in design
- Have many tables
- Have many databases

An Example Enterprise-level Database System



Commercial DBMS Products

- Example Desktop DBMS Products
  - Microsoft Access

- Example Organizational DBMS Products

  SQL Server

  Oracle

  MySQL

  DB2

Components of DBMS Environment

- **Hardware**
  - Can range from a PC to a network of computers.
- **Software**
  - DBMS, operating system, network software (if necessary) and also the application programs.
- **Data**
  - Used by the organization and a description of this data called the schema.
- **Procedures**
  - Instructions and rules that should be applied to the design and use of the database and DBMS.
- **People**

People Who Work with Databases

1. **Database implementers**
   - work for vendor such as IBM or Oracle.
2. **End users**
   - come from a diverse and increasing number of fields
   - Many end users simply use applications written by database application programmers
3. **Database application programmers**
   - develop packages that facilitate data access for end users, who are usually not computer professionals, using the host or data languages and software tools that DBMS vendors provide.
   - (Such tools include report writers, spreadsheets, statistical packages, and the like.)
   - Application programs should ideally access data through the external schema

People Who Work with Databases

4. Database Administrator

- The DBA is responsible for many critical tasks:

a) Design of the Conceptual and Physical Schemas:

— The DBA is responsible for interacting with the users of the system to understand what data is to be stored in the DBMS and how it is likely to be used.

— Based on this knowledge, the DBA must design the conceptual schema (decide what relations to store) and the physical schema (decide how to store them).

— The DBA may also design widely used portions of the external schema, although users probably augment this schema by creating additional views.

People Who Work with Databases

4. Database Administrator

b) Security and Authorization:

— The DBA is responsible for ensuring that unauthorized data access is not permitted.

— In general, not everyone should be able to access all the data. In a relational DBMS, users can be granted permission to access only certain views and relations.

c) Data Availability and Recovery from Failures:

— The DBA must take steps to ensure that if the system fails, users can continue to access as much of the uncorrupted data as possible.

— The DBA must also work to restore the data to a consistent state.

— The *DBMS provides software support for* these functions, but the DBA is responsible for implementing procedures to back up the data periodically and maintain logs of system activity (to facilitate recovery from a crash).

People Who Work with Databases

4. Database Administrator

d) Database Tuning:

- Users' needs are likely to evolve with time.
- The DBA is responsible for modifying the database, in particular the conceptual and physical schemas, to ensure adequate performance as requirements change.

Examples of Database Applications

- Purchases from the supermarket
- Purchases using your credit card
- Booking a holiday at the travel agents
- Using the local library
- Taking out insurance
- Renting a video
- Using the Internet
- Studying at university